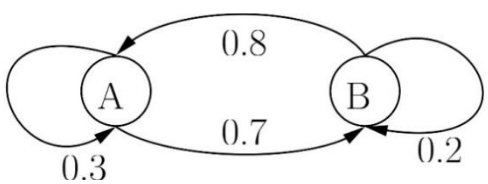
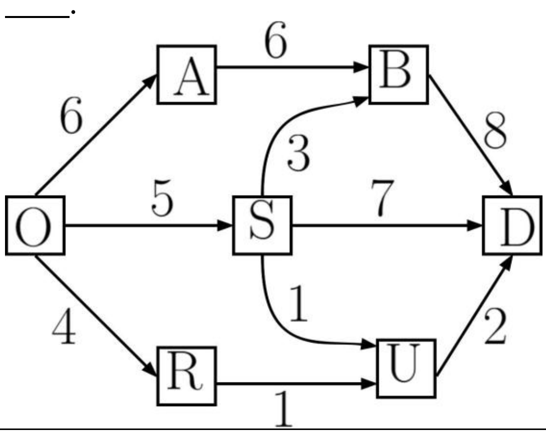
R. V. COLLEGE OF ENGINEERING®, Bengaluru-59
(Autonomous Institution under VTU)
VI Semester B. E. Examinations
Model Question Paper
Mathematical Modelling (MA266TEU)

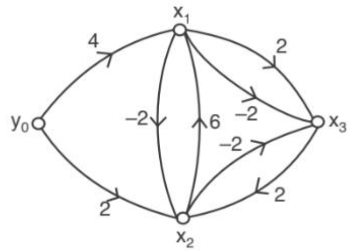
Time: 03 Hours ***Maximum Marks: 100***

Instructions to candidates:

1. **Answer all questions from Part A. Part A questions should be answered in first three pages of the answer book only.**
2. **Answer FIVE full questions from Part B.**

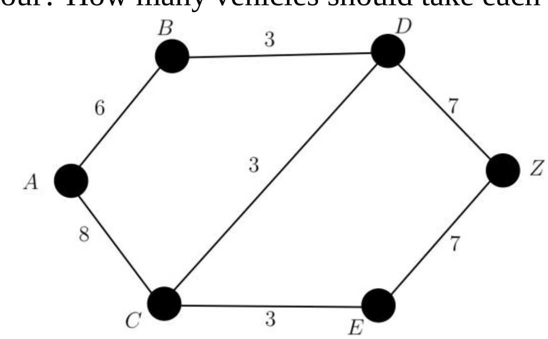
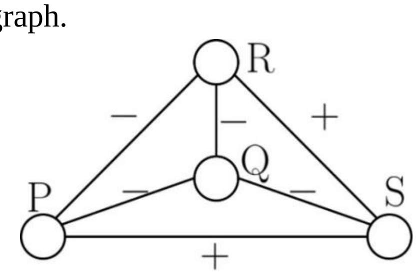
PART - A

Q.No		Question	M	BTL	CO
1	1.1	The population of California fox are decreasing at the rate of 5% per year. If their population was 400 in 2013, then how many of them will be left approximately in 2023?	2	2	1
	1.2	The growth of an organism follows the law $\frac{dx}{dt} = -\frac{1}{13}x$. If initially there were 200 of them, then after 13 years, their approximate number will be _____ and their half-life period is approximately _____ years.	2	2	3
	1.3	In terms of y_{t+1} and y_{t+2}, the arbitrary constant M in the equation $y_t = M2^t + N(-3)^t$ is equal to ______.	2	3	3
	1.4	The curve which extremizes the functional $\int_{10}^{11}[y^2 + x^2y']\,dx$ is ______.	2	2	2
	1.5	In the long run, the proportion of time that the given Markov chain will be in state A is _____ and state B is ______. 	2	2	2
	1.6	The angle of projection for a projectile to attain a) the maximum horizontal distance should be ______. b) the maximum height should be ______.	2	1	3
	1.7	In the following figure, with respect to dynamic programming, the value of $f_1^*(O)$ is ____. 	2	2	4

	1.8	Write the linear equations associated with the following signal-flow graph: 	2	2	4
	1.9	Given the initial state vector (1, 0) and the transition matrix shown below, find the state vector corresponding to two steps later (n = 2). $$\begin{bmatrix} 0.13 & 0.87 \\ 0.91 & 0.09 \end{bmatrix}$$	2	1	1
	1.10	‘Edge coloring’ is the assigning of colors to the edges so that no two adjacent edges have same color. Consider a simple connected graph G in which the number of vertices is equal to the number of edges, and every vertex has degree 2. What is the minimum number of colors required to ‘edge-color’ G?	2	1	2

PART – B

2	a	Give a note on the classification of mathematical models giving at least four examples.	8	1	1
	b	Drug is induced in a patient's bloodstream at regular intervals of duration T each. Simultaneously the drug disappears at a rate proportional to the concentration c(t) of the drug present at any time t. Determine the differential equation governing c(t), solve it and interpret the results.	8	3	3
3	a	Mathematically establish the Hardy-Weinberg law in genetics.	8	3	2
	b	A person takes a loan of ₹S_0 with i as the compound interest rate per month. If the loan is to be paid back in n equal instalments by paying ₹R per month, then show that $R = S_0 \frac{i}{1-(1+i)^{-n}}$.	8	2	2
		OR			
4	a	Bring out the dynamics of price and quantity of a commodity produced by a farm according to the Cobweb model.	8	3	2
	b	i. Solve the difference equation $x_{t+2} = 4(x_{t+1} - x_t), t \geq 0,\ x_0 = 1, x_1 = 3$. ii. The number of virus affected files in a system is 5000 (to start with) and this increases 200% every two hours. Model this using a difference equation and find the number of virus affected files in the system after one day.	8	2	2
5	a	Check if the transition probability matrix $\begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1/3 & 1/3 & 1/3 \end{bmatrix}$ is irreducible. Also find its stationary probability vector.	8	2	1
	b	A city is served by two cable TV companies, Best TV and Cable Cast. Due to their aggressive sales tactics, each year 40% of Best TV customers switch to Cable Cast. On the other hand, 30% of the Cable Cast customers switch to Best TV. If the initial market	8	2	3

		share for Best TV is 20% and for Cable Cast is 80%, find the market share of each company after 5 years.			
		OR			
6	a	Consider a network containing five individuals viz. a, b, c, d and e. The system a can communicate to only b and c and its chances of communicating with each are in the ratio 3:2 respectively. Similarly, b communicates with a and c in ratio 1:1: c can communicate with a, b and d in ratio $4:3:3$: d communicates with c and e in ratio $3:7$ and e can communicate to b only. Model the situation as a Markov chain and rank the five individuals in accordance with their importance in the network in the long run.	8	2	1
	b	A gambler has ₹2. He bets ₹1 at a time and wins ₹1 with probability $\frac{1}{2}$. He stops playing if he ends up with no money or wins ₹4. a. What is the TPM of the related Markov chain? b. What is the probability that he has lost his money at the end of 5 plays? What is the probability that the game lasts more than 7 plays?	8	3	3
7	a	At the end of the school day, all the students plan on driving from school (vertex A) to the concert (at vertex Z). The edges in the graph below represent one-way roads, and the weight of each edge represents the number of vehicles (in hundreds) that particular road can handle in one hour. What is the greatest number of vehicles that can get from school to the concert in one hour? How many vehicles should take each road? 	8	2	4
	b	i. Basavaraj has picked 8 friends for a project team such that they are all friends among themselves. Considering A and B being friends as a single friendship, is it possible that there are totally 38 friendships? Substantiate your answer. ii. Define the degree of balance of a signed graph. The following graph models the social relation between 4 people with '+' indicating friendship and '–' indicating enmity. Find the degree of balance of the graph. 	8	3	4
		OR			
8	a	The Fig 8b shows an intersection of two streets where there is often heavy traffic. There	8	2	4

<table>
<tr><td></td><td></td><td>are seven traffic lanes $L1, L2, \dots L7$ where vehicles can enter the intersection of these two streets. A traffic light is located at this intersection. During a certain phase of this traffic light, those cars in lanes for which the light is green may proceed safely through the intersection. Model this situation by a graph giving out the details as to what the vertices and the edges represent. Use that graph to determine whether it is possible with four phases, for cars in all lanes to proceed safely through the intersection.

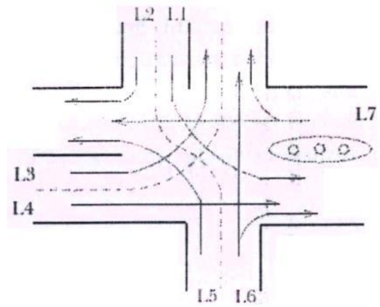



Fig 8b</td><td></td><td></td><td></td></tr>
<tr><td></td><td>b</td><td>Define the Harary measure of status. Find the Harary measure of each individual in the organizational graph shown in Fig 9c.

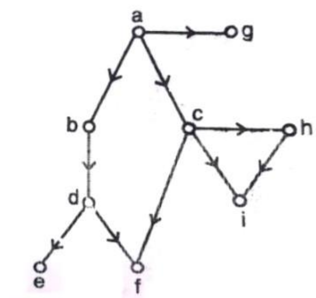



Fig 9c</td><td>8</td><td>3</td><td>4</td></tr>
<tr><td>9</td><td>a</td><td>A person decides to travel from a place A to a place J. The journey would require traveling by a stagecoach through different cities on the paths of which there is danger of attack by thieves. Life insurance policies were offered to the passengers, the safest route would be the one with the cheapest total life insurance policy. The details of the insurance from one place to another are given below. Model it through a directed graph and find the route which minimizes the cost using dynamic programming.
<table><tr><th></th><th>B</th><th>C</th><th>D</th></tr><tr><td>A</td><td>1</td><td>3</td><td>6</td></tr></table>
<table><tr><th></th><th>E</th><th>F</th><th>G</th></tr><tr><td>B</td><td>5</td><td>4</td><td>6</td></tr><tr><td>C</td><td>4</td><td>3</td><td>2</td></tr><tr><td>D</td><td>1</td><td>4</td><td>5</td></tr></table>
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</td><td>8</td><td>1</td><td>2</td></tr>
<tr><td></td><td>b</td><td>Model the problem of light passing from a point A to a point B as a variational problem by applying Fermat’s principle of least time. Show that a consequence of this modelling is the Snell’s law of refraction.</td><td>8</td><td>3</td><td>4</td></tr>
<tr><td></td><td></td><td>OR</td><td></td><td></td><td></td></tr>
<tr><td>10</td><td>a</td><td>Discuss any five optimization techniques and any three optimization principles.</td><td>8</td><td>1</td><td>2</td></tr>
<tr><td></td><td>b</td><td>A particle descends from one point to another point in the absence of friction under the action of gravity. Model the situation as a variational problem so as to find the path of the quickest descent.</td><td>8</td><td>3</td><td>4</td></tr>
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